PORTALDOT HACKER HOUSE 2026 · TRACK: NODE OPERATIONS AND TOOLING

# NodeScope

## Node truth suite for Portaldot V3

Two pillars. One evidence discipline.

HEALTH FEES

nodescope.vercel.app · github.com/Firzadf/nodescope

dorahacks.io/build/50146 · Demo video: youtu.be/TyTudF3MUaI

PROBLEM

# A node can answer once. That proves almost nothing.

### Health over time

An operator can see a node respond, but cannot prove it stayed healthy, or what changed since the last check.

### One operation, two prices

The same operation is priced on the EVM surface and on the native surface, and headline fee numbers hide the difference.

SOLUTION

# One build, two pillars, one evidence discipline.

### Health

Timestamped snapshots. Deterministic before/after comparison. Reports verified by SHA-256.

### Fees

One operation priced on EVM and native, side by side. Fee-only and all-in shown as they are.

**No background polling. No black-box score. No simulated data shown as live.**

DEMO FLOW · 60 SECONDS

# From endpoint to verified report in five steps.

**01** Open /app. The testnet endpoint is already filled in.

**02** Run full diagnostic: connect, health check, capability discovery, Substrate probe, snapshot, combined verdict.

**03** Read the verdict bar: overall verdict, checks passed, WSS status, latency, last-checked time.

**04** Compare two snapshots: new and resolved findings, block delta, latency trend.

**05** Export the report and verify it at /verify. No account. No wallet.

ON-CHAIN EVIDENCE · PORTALDOT V3 TESTNET · CHAIN 420420777

# Three reports anchored. Anyone can recheck them.

**Registry** 0xd48da7eff3264bdd51a86b873da9ff6997aae99c

**Deploy tx** 0x1c2a055523a4cf8b73d771df4d3c54d21392a85f86a14a4d68213bf6db5f4904

Ownerless contract · Deploy block 216268

## Health #1

Anchor block 216341

**Report** 0x08ac72ad0f7a818bc18a8624232811fc4f994cf065bbd2cad1eea4b0bb002df4

**Tx** 0xba35df8e2f182d75885b75b75f0dd97340f4ce64333840d1cf06a252441e9476

## Fees

Anchor block 216343

**Report** 0x246d6d43b308b8801109910bb52d8ad8d264376e7c8557255b3704176ada0464

**Tx** 0x0c08335507cfb2a0df4b0236543607aed04527ec26361e6e0022c9100c88f2c9

## Health #2

Anchor block 216346

**Report** 0x528f38d58920b2c7fef730c27e95200e429a126874746e8cead5c1146c768a5b

**Tx** 0xc74a08f3835b0c5b7d813ab15597488affa1987bce79a88d843dfdc31a9ec913

Check any anchor at nodescope.vercel.app/verify

FEES FINDING · MEASURED 27 OCT 2026 · CONTRACT DEPLOY

# The headline says 804x. The bill says 1.016x.

**804x**

fee-only

**1.016x**

all-in

EVM deploy: 124.098 tPOT · Native all-in: 122.191 tPOT · Within 2%

Most of the gap is a 122.037 tPOT storage deposit on the native side. A deposit is a hold, not a burned fee: it is refundable when storage is freed. The EVM figure already folds it in.

tPOT is a testnet token with no market price.

ENGINEERING

# Built to be rechecked, not to be believed.

| | |
|---|---|
| **470 tests across 68 files** | Vitest, run on every push. |
| **5/5 Foundry contract tests** | The attestation contract, tested in isolation. |
| **CI on Ubuntu and Windows** | Node on both; Foundry on Ubuntu. |
| **CLI and GitHub Action** | check and fees without the browser; a pre-deploy cost gate teams can run. |
| **Deterministic comparison** | Same inputs, same diff. Reports verified by SHA-256. |
| **Read-only by design** | No wallet, no signing, no background polling. |

ECOSYSTEM CONTRIBUTION

# Findings the ecosystem can use, not just a demo.

· Six measured findings for Portaldot, documented with reproduction steps: fee model, deposit semantics, dry-run limits, WSS connect, unavailable equivalents.

· A GitHub Action any team can run as a pre-deploy cost gate.

· An evidence standard: anchored reports anyone can recompute from the repo.

· Adoption outreach prepared. No team results claimed yet.

BUSINESS MODEL

# Free to use. Paid when it runs for you.

### Free, open source (MIT)

The app, Fee Lab, CLI, GitHub Action, and the attestation registry.

### Paid, when there is demand

Hosted monitoring with alerts for node operators. A managed pre-deploy cost gate with history for teams and CI.

Targets: Portaldot builders, node operators, wallets and dApps that need EVM vs native routing.

Sizing, honestly: no total market claimed. Only verified counts are used.

ROADMAP · TEAM

# What happens next.

| | |
|---|---|
| **Hosted monitoring and alerts** | For node operators, when demand exists. |
| **Pre-deploy cost gate in CI** | The GitHub Action as a check teams run before deploying. |
| **Mainnet check** | The mainnet profile becomes a live check when a public mainnet EVM RPC is verified. |
| **GitHub Action on Marketplace** | Publish the action so any team can add it in one step. |

**Team: Firza Duta Fauzan, solo builder.**

**Verify it yourself: nodescope.vercel.app/verify**